

Group Project: End-to-End NLP Pipeline: From Data to Insights and Retrieval

Learning Objectives

By completing this project, students will be able to:

1. Design and implement an end-to-end NLP pipeline that integrates data preprocessing, modeling, and information retrieval for real-world text analytics.
2. Apply advanced NLP techniques selectively by choosing and justifying one analysis track (Entity Extraction or Sentiment Analysis) and one modeling track (Text Classification or Topic Modeling).
3. Evaluate and compare models using appropriate metrics, perform error analysis, and interpret results to assess performance and limitations.
4. Communicate technical findings effectively in a professional report and presentation, including visualizations, insights, and ethical considerations.

Suggested Domains & Datasets

Choose **one** dataset that can support the selected tracks and BM25 retrieval.

1. News datasets

- [BBC News Summary](#) (Kaggle): Short news articles across categories like politics, tech, sports.
- [Reuters-21578](#): Classic dataset for text classification and topic modeling.
- [AG News Corpus](#): Large-scale news dataset suitable for classification and IR.

Why choose this domain?

Rich named entities (people, organizations, locations) for NER; clear topical structure for topic modeling.

2. Reviews datasets

- [Amazon Reviews](#) : Millions of product reviews with ratings and categories.
- [Amazon Fine Food Reviews](#)

Why choose this domain?

Ideal for sentiment analysis; categories enable classification; topics reveal customer concerns.

3. Social Media

- [Disaster tweets](#)
- [Social media sentiments](#)

Why choose this domain?

Short, noisy text for real-world challenges; strong sentiment signal; hashtags and mentions for NER.

4. Medical & Healthcare

- [Drug Review Dataset](#) (UCI): Patient reviews of medications with ratings and conditions.

If using social media, include steps for **de-duplication**, **spam removal**, **language filtering**.

5. Spam & Email Filtering

- [SMS Spam Collection](#) (UCI): Labeled SMS messages as spam or ham.
- [Enron Email Dataset](#): Large corpus of emails for spam detection and topic modeling.

Why choose this domain?

Perfect for text classification tasks; spam detection is a classic NLP application; IR can be applied for searching relevant emails.

Project Components

Part A: Data Loading & Preprocessing (20%)

- Data ingestion + schema description.
- Cleaning: lowercasing, punctuation/URL removal (justify choices), handling emojis if social media.
- Tokenization, stopwords, lemmatization/stemming (if applicable).
- Exploratory analysis: length distributions, class imbalance, vocabulary.

Part B: Analysis Track - Select ONE (Entity Extraction/ Sentiment Analysis) (20%)

B1) Entity Extraction (NER)

- **Baseline:** spaCy pretrained NER (en_core_web_lg) or HuggingFace token-classification model.
- **Evaluation:** Precision, Recall, F1 (entity-level, strict and partial matches); error breakdown by entity type; calibration plot if using confidence.

B2) Sentiment Analysis

- Lexicon-based baseline (e.g., VADER/AFINN) with domain adjustments (negation handling, intensifiers).
- Supervised model: traditional (LogReg + TF-IDF) vs. fine-tuned transformer (DistilBERT/BERT).
- Address class imbalance (class weights).
- **Evaluation:** Accuracy, Macro-F1, per-class F1.

Part C: Modeling Track : Select ONE (Text Classification/ Topic Modeling) (30%)

C1) Text Classification

- Baselines: Multinomial Naive Bayes / Logistic Regression with bag-of-words, TF-IDF.
- Advanced: CNN/RNN or transformer fine-tuning;
- **Evaluation:** Macro-F1, per-class F1, confusion matrix, class-wise PR curves; robustness across document length.

C2) Topic Modeling (LDA & NMF)

- Implement both **LDA** and **NMF**.
- **Topic number selection:** grid search with coherence (C_v) and intruder-word test (human evaluation optional).
- **Stability:** run seeds; compare topic consistency.
- **Evaluation:** Topic coherence, diversity, sparsity; qualitative labeling of topics;

Deliverables: pyLDavis (or equivalent), top terms per topic, representative documents, topic labeling rationale, comparison table LDA vs NMF.

Part D: Information Retrieval (Required) — BM25(20%)

- Build a BM25-based search engine over the corpus (document-level).
- Query interface: keyword queries; optional query expansion (synonyms, pseudo-relevance feedback).
- **Evaluation:**
 - **Precision@k, Recall@k, MAP, MRR, nDCG@k.**

- Construct a **test query set** with relevance judgments (manual or heuristic).
- Error analysis: query drift, stopwords effects, term mismatch.

Deliverables: retrieval demo notebook or minimal app, metrics dashboard, sample queries & relevance judgments.

Part E: Presentation and report quality(10%)

Submission

Deliverables: jupyter notebook for each section.

Due date [Insert Date]